DSC 40B

Theoretical Foundations II

Lecture 6 | Part 1

Solving Recurrences

Today in DSC 40B...

- Finish Lecture 05: solving recurrences, and the time complexity of binary search.
- Then: sorting.

Last Time: Getting Rich

- You start a job. On day 1, you are paid $3.
- Each day thereafter, your pay doubles.
- **Problem:** How much will you be paid on day n?

Last Time: Getting Rich

- We derived a **recurrence relation** for $S(n)$, your pay on day n:

$$S(n) = \begin{cases} 2 \cdot S(n-1), & n \geq 2 \\ 3, & n = 1 \end{cases}$$

- Now we want to **solve** it: find a simple, **non-recursive** formula for $S(n)$.

Solving Recurrences

We'll use a four-step process to solve recurrences:

1. "Unroll" several times to find a pattern.
2. Write general formula for *k*th unroll.
3. Solve for # of unrolls needed to reach base case.
4. Plug this number into general formula.

Step 1: Unroll several times

$$S(n) = \begin{cases} 2 \cdot S(n-1), & n \geq 2 \\ 3, & n = 1 \end{cases}$$

$S(n) = 2\,S(n-1)$ $\quad k=1$

$= 2\,[2\,S(n-2)]$

$= 4\,S(n-2)$ $\quad k=2$

$= 4\,[2\,S(n-3)]$

$= 8\,S(n-3)$ $\quad k=3$

$S(n-1) = 2\,S(n-2)$

$S(n-2) = 2\,S(n-3)$

Step 2: Find general formula

$$
\begin{aligned}
S(n) &= 2 \cdot S(n-1) && k=1 \\
&= 2 \cdot 2 \cdot S(n-2) && k=2 \\
&= 2 \cdot 2 \cdot 2 \cdot S(n-3) && k=3
\end{aligned}
$$

On step k:

$$S(n) = 2^k S(n-k)$$

Step 3: Find step # of base case

- On step k, $S(n) = 2^k \cdot S(n - k)$.
- When do we see $S(1)$?

Solve: $n - k = 1$? $\rightarrow$ $k = n - 1$

Step 4: Plug into general formula

- From step 2: $S(n) = 2^k \cdot S(n - k)$.
- From step 3: Base case of $S(1)$ reached when $k = n - 1$.
- So: $S(n) = 2^{n-1} S(n-(n-1))$

$= 2^{n-1} S(1)$

$= 2^{n-1} \times 3$

Solving the Recurrence

$2^{n-1} = 2^n \cdot 2^1$

- We have **solved** the recurrence[1]:

$$S(n) = 3 \cdot 2^{n-1}$$

- This is the **exact** solution. The **asymptotic** solution is $S(n) = \Theta(2^n)$.

- We'll call this method "solving by unrolling".

[1] On day 20, you'll be paid ≈1.5 million dollars.

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Lecture 6 | Part 2

Binary Search Recurrence

Last Time: Binary Search

- What is the time complexity of `binary_search`?

```
import math
def binary_search(arr, t, start, stop):
    """
    Searches arr[start:stop] for t.
    Assumes arr is sorted.
    """
    if stop - start <= 0:
        return None
    middle = math.floor((start + stop)/2)
    if arr[middle] == t:
        return middle
    elif arr[middle] > t:
        return binary_search(arr, t, start, middle)
    else:
        return binary_search(arr, t, middle+1, stop)
```

$T(n) = c + T(n/2)$

Binary Search

- Best case: $\Theta(1)$.
- For the worst case, we wrote down a recurrence relation:

$$T(n) = \begin{cases} T(n/2) + \Theta(1), & n \geq 2 \\ \Theta(1), & n = 1 \end{cases}$$

Simplification

- When solving, we can replace $\Theta(f(n))$ with $f(n)$:

$$T(n) = \begin{cases} T(n/2) + 1, & n \geq 2 \\ 1, & n = 1 \end{cases}$$

- As long as we state final answer using Θ notation!

Another Simplification

- When solving, we can assume n is a power of 2.

Step 1: Unroll several times

$$T(n) = \begin{cases} T(n/2) + 1, & n \geq 2 \\ 1, & n = 1 \end{cases}$$

$T(n) = T(n/2) + 1$ $\quad k=1$

$= [T(n/4) + 1] + 1$ $\quad k=2$

$= T(n/4) + 2$

$= [T(n/8) + 1] + 2$

$= T(n/8) + 3$ $\quad k=3$

$T\left(\frac{n}{2}\right) = T\left(\frac{(n/2)}{2}\right) + 1$

$= T\left(\frac{n}{4}\right) + 1$

$T(n/4) = T(n/8) + 1$

Step 2: Find general formula

$$T(n) = T(n/2) + 1 \qquad k=1$$
$$= T(n/4) + 2 \qquad k=2$$
$$= T(n/8) + 3 \qquad k=3$$

On step k: $T(n) = T(n/2^k) + k$

Step 3: Find step # of base case

- On step k, $T(n) = T(n/2^k) + k$
- When do we see $T(1)$?

Solve: $\frac{n}{2^k} = 1$ for k: $k = \log_2 n$

Step 4: Plug into general formula

- $T(n) = T(n/2^k) + k$
- Base case of $T(1)$ reached when $k = \log_2 n$.
- So:

$$T(n) = T\left(\frac{n}{2^{\log_2 n}}\right) + \log_2 n$$

($k = \log_2 n$)

$$= T\left(\frac{n}{n}\right) + \log_2 n = T(1) + \log_2 n$$

($T(1) = \Theta(1)$)

$$= \log_2 n + \Theta(1)$$

$$= \Theta(\log n)$$

Note

- Remember: $\log_b x = (\log_a x)/(\log_a b)$
- So we don't write $\Theta(\log_2 n)$
- Instead, just: $\Theta(\log n)$

Time Complexity of Binary Search

- Best case: $\Theta(1)$
- Worst case: $\Theta(\log n)$

Is binary search fast?

- Suppose all 10^{19} grains of sand are assigned a unique number, sorted from least to greatest.
- Goal: find a particular grain.
- Assume one basic operation takes 1 nanosecond.

Is binary search fast?

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- Assume one basic operation takes 1 nanosecond.
- Linear search: 317 years.

Is binary search fast?

- Suppose all 10^{19} grains of sand are assigned a unique number, sorted from least to greatest.
- Goal: find a particular grain.
- Assume one basic operation takes 1 nanosecond.
- Linear search: 317 years.
- Binary search: ≈ 60 nanoseconds.

Exercise

Binary search seems so much faster than linear search. What's the caveat?

Caveat

- The array must be **sorted**.
- This takes $\Omega(n)$ time.

Why use binary search?

- If data is **not sorted**, sorting + binary search takes longer than linear search.
- But if doing **multiple queries**, looking for nearby elements, sort once and use binary search after.

Live Q&A

Exercise

Suppose `is_sorted` has the best possible time complexity. What is the **worst case** time complexity of `careful_binary_search`?

A) $\Theta(1)$ B) $\Theta(\log n)$ C) $\Theta(n)$ D) $\Theta(n \log n)$

```
def careful_binary_search(arr, t):
    if not is_sorted(arr):
        raise ValueError("arr must be sorted")
    return binary_search(arr, t, 0, len(arr))
```

Theoretical Lower Bounds

- A tight lower bound for searching a sorted list is $\Omega(\log n)$.
- This means that binary search has **optimal** worst case time complexity.

Databases

- Some database servers will **sort** by key, use binary search for queries.
- Often instead of sorting, **B-Tree indexes** are used.
- But sorting + binary search still used when space is limited.

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Theoretical Foundations II

Lecture 6 | Part 3

Selection Sort and Loop Invariants

Sorting

- Sorting is a very common operation.
- But why is it important?

Sorting

- Sorting is a very common operation.
- But why is it important?
- `A e s t h e t i c` reasons?

Sorting

- Sorting is a very common operation.
- But why is it important?
- A e s t h e t i c reasons?
- Sorting makes some problems easier to solve.

Today

- How do we sort?
- How fast can we sort?
- How do we use sorted structure to write faster algorithms?

Today

- **Also:** how to understand complex loops with **loop invariants**.

Selection Sort

- Repeatedly remove smallest element.
- Put it at beginning of new list.

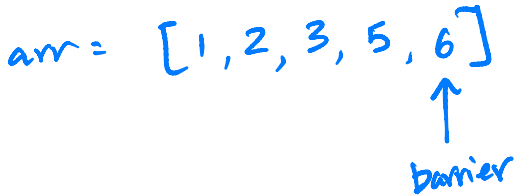
Example: `arr = [5, 6, 3, 2, 1]`

output = [1, 2, 3, 5, 6]

In-place Selection Sort

- We don’t need a separate list.
  - We can swap elements until sorted.
- Store “new” list at the beginning of input list.
- Separate the old and new with a **barrier**.

Example: `arr = [5, 6, 3, 2, 1]`



a, b = b, a

```
def selection_sort(arr):
    """In-place selection sort."""
    n = len(arr)
    if n <= 1:
        return
    for barrier_ix in range(n-1):
        # find index of min in arr[start:]
        min_ix = find_minimum(arr, start=barrier_ix)
        #swap
        arr[barrier_ix], arr[min_ix] = (
                arr[min_ix], arr[barrier_ix]
        )
```

```
def find_minimum(arr, start):
    """Finds index of minimum. Assumes non-empty."""
    n = len(arr)
    min_value = arr[start]
    min_ix = start
    for i in range(start + 1, n):
        if arr[i] < min_value:
            min_value = arr[i]
            min_ix = i
    return min_ix
```

Loop Invariants

- How do we understand an iterative algorithm?
- A **loop invariant** is a statement that is true after every iteration.
  - And before the loop begins!

Loop Invariant(s)

After the αth iteration of selection sort, each of the first α elements is $\leq$ each of the remaining elements.

Example: `arr = [5, 6, 3, 2, 1]`

Loop Invariant(s)

After the αth iteration, the first α elements are sorted.

Example: `arr = [5, 6, 3, 2, 1]`

Loop Invariants

- Plug the total number of iterations into the loop invariant to learn about the result.
  - `selection_sort` makes $n - 1$ iterations:
  - After the $(n - 1)$th iteration, the first $(n - 1)$ elements are sorted.
  - After the $(n - 1)$th iteration, each of the first $(n - 1)$ elements is $\leq$ each of the remaining elements.

Time Complexity

$\Theta(n^2)$

```
def selection_sort(arr):
    """In-place selection sort."""
    n = len(arr)
    if n <= 1:
        return
    for barrier_ix in range(n-1):
        # find index of min in arr[barrier_ix:]
        min_value = arr[barrier_ix]
        min_ix = barrier_ix
        for i in range(barrier_ix + 1, n):
            if arr[i] < min_value:
                min_value = arr[i]
                min_ix = i
        #swap
        arr[barrier_ix], arr[min_ix] = (
                arr[min_ix], arr[barrier_ix]
        )
```

Time Complexity

- Selection sort takes $\Theta(n^2)$ time.

Exercise

Modify `selection_sort` so that it computes a **median** of the input array. What is the time complexity?

```
def selection_sort(arr):
    """In-place selection sort."""
    n = len(arr)
    if n <= 1:
        return
    for barrier_ix in range(n-1):
        # find index of min in arr[start:]
        min_ix = find_minimum(arr, start=barrier_ix)
        #swap
        arr[barrier_ix], arr[min_ix] = (
                arr[min_ix], arr[barrier_ix]
        )
```

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Theoretical Foundations II

Lecture 6 | Part 4

Mergesort

Can we sort faster?

- The tight theoretical lower bound for **comparison** sorting is $\Theta(n \log n)$.
- Selection sort is quadratic.
- How do we sort in $\Theta(n \log n)$ time?

Mergesort

- Mergesort is a fast sorting algorithm.
- Has **best possible** (worst-case) time complexity: $\Theta(n \log n)$.
- Implements **divide/conquer/recombine** strategy.

The Idea

- **Divide**: split the array into halves
  - `[6,1,9,2,4,3]` → `[6,1,9]`, `[2,4,3]`
- **Conquer**: sort each half, recursively
  - `[6,1,9]` → `[1,6,9]` and `[2,4,3]` → `[2,3,4]`
- **Combine**: merge sorted halves together
  - `[1,6,9]`, `[2,3,4]` → `[1,2,3,4,6,9]`

Aside: splitting arrays

- Splitting an array in half by **slicing**:

```
>>> arr = [9, 1, 4, 2, 5]
>>> middle = math.floor(len(arr) / 2)
>>> arr[:middle]
[9, 1]
>>> arr[middle:]
[4, 2, 5]
```

middle

0

$\Theta(n)$ time!

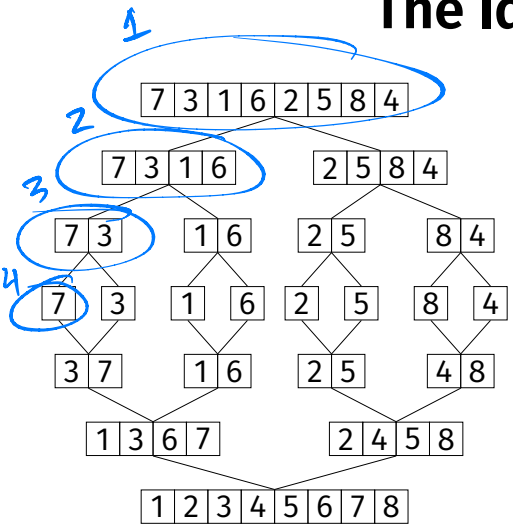
- **Warning!** Creates a copy!

Mergesort

```
def mergesort(arr):
    """Sort array in-place."""
    if len(arr) > 1:
        middle = math.floor(len(arr) / 2)
        left = arr[:middle]
        right = arr[middle:]
        mergesort(left)
        mergesort(right)
        merge(left, right, arr)
```

print(arr)

The Idea


1
2
3
4
7 3 1 6 2 5 8 4
7 3 1 6
2 5 8 4
7 3
1 6
2 5
8 4
7
3
1
6
2
5
8
4
3 7
1 6
2 5
4 8
1 3 6 7
2 4 5 8
1 2 3 4 5 6 7 8

Understanding Mergesort

1. What is the base case?
2. Are the recursive problems smaller?
3. Assuming the recursive calls work, does the whole algorithm work?

1. Base Case: $n = 1$

- Arrays of size one are trivially sorted.
- Returns immediately. **Correct!**

2. Smaller Problems?

- Are `arr[:middle]` and `arr[middle:]` always smaller than `arr`?
- Try it for `len(arr) == 2`.

3. Does it Work?

- Assume `mergesort` works on arrays of size $< n$.
- Does it work on arrays of size n?

Mergesort

```
def mergesort(arr):
    """Sort array in-place."""
    if len(arr) > 1:
        middle = math.floor(len(arr) / 2)
        left = arr[:middle]
        right = arr[middle:]
        mergesort(left)
        mergesort(right)
        merge(left, right, arr)
```

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Theoretical Foundations II

Lecture 6 | Part 5

Merge

Merging

- We have sorted each half.
- Now we need to **merge** together.

Merging

- We have sorted each half.
- Now we need to **merge** together.
- **Note:** this is an example of a problem that is made easier by sorting.

merge

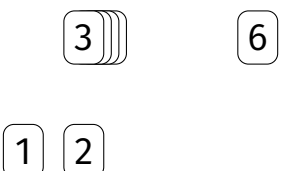




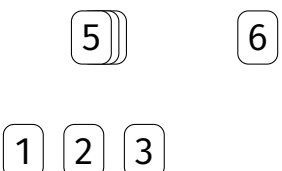
merge



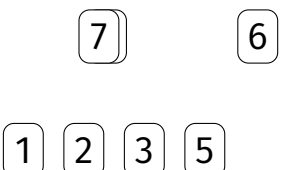
merge



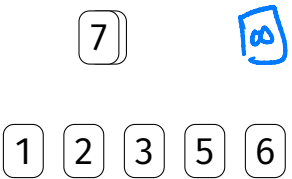
merge



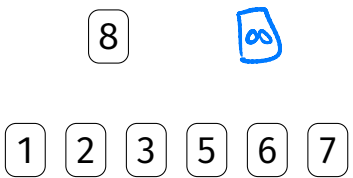
merge



merge



merge



merge



merge

```
def merge(left, right, out):
    """Merge sorted arrays, store in out."""
    left.append(float('inf'))
    right.append(float('inf'))
    left_ix = 0
    right_ix = 0

    for ix in range(len(out)):
        if left[left_ix] < right[right_ix]:
            out[ix] = left[left_ix]
            left_ix += 1
        else:
            out[ix] = right[right_ix]
            right_ix += 1
```

Loop Invariant

- Assume `left` and `right` are sorted.
- **Loop invariant**: After αth iteration, first α elements of `out` are the smallest α elements of those in `left` and `right`, in sorted order.
- That is, after αth iteration,
  `out[:`α`] == sorted(left + right)[:`α`]`

Key of `mergesort`

- `merge` is where the **actual sorting** happens.
- Example: `merge([3], [1], ...)` results in `[1,3]`

Time Complexity of `merge`

```
def merge(left, right, out):
    """Merge sorted arrays, store in out."""
    left.append(float('inf'))
    right.append(float('inf'))
    left_ix = 0
    right_ix = 0

    for ix in range(len(out)):
        if left[left_ix] < right[right_ix]:
            out[ix] = left[left_ix]
            left_ix += 1
        else:
            out[ix] = right[right_ix]
            right_ix += 1
```

n

O(n)

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Theoretical Foundations II

Lecture 6 | Part 6

Time Complexity of Mergesort

Time Complexity

```
def mergesort(arr):
    """Sort array in-place."""
    if len(arr) > 1:
        middle = math.floor(len(arr) / 2)
        left = arr[:middle]
        right = arr[middle:]
        mergesort(left)
        mergesort(right)
        merge(left, right, arr)
```

Aside: Copying

- What is `arr[:middle]` doing “under the hood”?
- What is the time complexity?

The Recurrence

```
def mergesort(arr):
    """Sort array in-place."""
    if len(arr) > 1:
        middle = math.floor(len(arr) / 2)
        left = arr[:middle]
        right = arr[middle:]
        mergesort(left)
        mergesort(right)
        merge(left, right, arr)
```

Solving the Recurrence

$$T(n) = 2T(n/2) + \Theta(n)$$

Optimality

- **Theorem**: Any (comparison) sorting algorithm's worst-case time complexity must be $\Omega(n \log n)$.
- Mergesort is **optimal**!

Be Careful!

- It is possible for a sorting algorithm to have a **best case** time complexity smaller than $n \log n$.
  - Insertion sort, for example.
- Mergesort has best case time complexity of $\Theta(n \log n)$.
- Mergesort is **sub-optimal** in this sense!

Be Careful!

- The $\Theta(n \log n)$ lower-bound is for **comparison sorting**.
- It is possible to sort in worst-case $\Theta(n)$ time without comparing.[2]

[2] Bucket sort, radix sort, etc.

What if?

- **Divide**: split the array into halves
- **Conquer**: sort each half **using selection sort**
- **Combine**: merge sorted halves together

Exercise

What is the time complexity of the code below?

A) $\Theta(n)$ B) $\Theta(n \log n)$ C) $\Theta(n^2)$ D) $\Theta(n^2 \log n)$

```
def mergeselectionsort(arr):
    """Sort array in-place."""
    if len(arr) > 1:
        middle = math.floor(len(arr) / 2)
        left = arr[:middle]
        right = arr[middle:]
        selection_sort(left)
        selection_sort(right)
        merge(left, right, arr)
```

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Theoretical Foundations II

Lecture 6 | Part 7

Using Sorted Structure

Sorted structure is useful

- Some problems become **much easier** if input is sorted.
  - For example, median, minimum, maximum.
- Sorting is useful as a **preprocessing** step.

Recall: The Movie Problem

- You're on a flight that will last *D* minutes.
- You want to pick two movies to watch.
- You want the total time of the two movies to be **as close as possible** to *D*.

The Movie Problem

- Brute force algorithm: $\Theta(n^2)$
- We can do better, if movie times are **sorted.**

Example

- Flight duration $D = 155$
- Movie times: $60, 80, 90, 120, 130$

	60	80	90	120	130
60					
80					
90					
120					
130					

Best pair:

The Algorithm

- Keep index of shortest and longest remaining.
- On every iteration, pair the shortest and longest.
- If this pair is too long, remove longest movie; otherwise remove shortest.
  - If times are **sorted**, finding new longest/shortest movie takes $\Theta(1)$ time!

60, 80, 90, 120, 130

The Algorithm

```
def optimize_entertainment(times, target):
    """assume times is sorted."""
    shortest = 0
    longest = len(times) - 1

    best_pair = (shortest, longest)
    best_objective = float('inf')

    for i in range(len(times) - 1):
        total_time = times[shortest] + times[longest]

        if abs(total_time - target) < best_objective:
            best_objective = abs(total_time - target)
            best_pair = (shortest, longest)

        if total_time == target:
            return (shortest, longest)
        elif total_time < target:
            shortest += 1
        else: # total_time > target
            longest -= 1

    return best_pair
```

Main Idea

Sorted structure allows you to rule out possibilities without explicitly checking them. But, it requires you to spend the time sorting first.

Tip: when designing an algorithm, think about sorting the input first.